Municipal air-quality decision-support pipeline

Batyrova, Yerkezhan | Matriculation no. 92131651 | DLBDSEDE02 Project: Data Engineering | Portfolio part 3 - Abstract

Purpose and stakeholder needs

The purpose of this project is to provide a reliable data layer for municipal air-quality decisions. The design begins with people who will actually use its outputs. Environmental analysts need raw measurements, missing-value information and warning labels, so they can decide whether a trend is trustworthy. City planners and policy teams need comparisons between stations and time periods when they discuss traffic or land-use measures. A dashboard developer needs recent readings and daily series in a stable format. The data steward needs run status, rejected rows and clear reconciliation figures. An authorised warning officer may review threshold cases, but the prototype cannot send messages to citizens. Such action requires approved policy and named responsibility.

These expectations were converted into testable conditions. Every source row must be either stored or quarantined, a repeated run must not create duplicates, and every ingestion run must finish with a visible status. The system also needs to return four representative queries within 2,000 ms on the declared local computer. This target is not a promise for every infrastructure. It is a practical prototype limit, and the benchmark records the measured time for later comparison.

Dataset limitations and possible bias

The UCI Beijing Multi-Site Air Quality dataset contains 420,768 hourly records from 12 stations for 2013-2017 (UCI Machine Learning Repository, n.d.). In this project it represents a future municipal batch feed and allows realistic volume testing. However, it is historical Beijing data and should not be used for conclusions about another city. The CSV files do not prove calibration accuracy, sensor maintenance, station placement or spatial coverage. They also do not show whether nearby stations observe the same local conditions. Even if the source was published by a government organisation, this does not automatically remove bias. Hourly measurements are suitable for many trends, but they may be too broad for short pollution events or health indicators based on another averaging period (World Health Organization, 2021).

The pipeline does not hide these issues. Missing measurements remain absent rather than being changed to zero. Configurable plausibility ranges are used only for technical screening and are not presented as medical limits. A doubtful but readable record is kept with a warning, while a structurally invalid row goes to quarantine with a reason. Query results include observation counts, which helps users see when two comparisons are based on unequal evidence. Before production use, municipality should add a station register with coordinates, sensor models, calibration dates and maintenance history.

Municipal air-quality decision-support pipeline

Batyrova, Yerkezhan | Matriculation no. 92131651 | DLBDSEDE02 Project: Data Engineering | Portfolio part 3 - Abstract

Technical implementation

The Python loader downloads the official source, opens its nested ZIP archive and reads the 12 station files in sequence. Every CSV row becomes one timezone-aware station-hour document. The ID is produced from station and timestamp, therefore the same row receives the same key during each run. Insert-only upserts make replay safe, while batches of 1,000 and limited exponential retries deal with temporary database problems. MongoDB validation protects the identity, time, provenance and quality fields, although the measurements object stays flexible for a future sensor metric (MongoDB, n.d.-a).

MongoDB was chosen because it supports indexed station and time queries while accepting a changing set of measurements. PostgreSQL would be a good option for a fixed relational registry, but plain files cannot provide the same concurrent indexed access. A warehouse would create unnecessary complexity for this stage. Separate collections store readings, rejected rows, ingestion runs, query benchmarks and source limitations. Docker Compose makes the prototype reproducible. A real deployment would still require secure secrets, TLS, backups, replication and workload testing before sharding.

Testing and observed results

Eleven unit tests passed and covered normalisation, missing values, new metrics, batching, nested archives, quality flags and user query contracts. The full Docker run processed all 420,768 rows. It stored 383,964 accepted observations and 36,804 warning observations, with zero structural rejections. Among the warnings, 36,788 records missed at least one required pollutant and 16 contained a value outside the screening range; these categories can overlap. Replay protection and run counters are available for further checks.

All four benchmark queries met the 2,000 ms target on the test laptop. The latest-reading query took 15.931 ms, the daily PM2.5 trend took 16.505 ms, the station comparison took 21.217 ms, and the quality summary took 221.352 ms. These results show that the prototype can deliver the required amount of data in reasonable time for one local run, but they do not replace concurrent production testing.

Reflection and next steps

The most important change after feedback was moving from a developer-centred idea to a stakeholder view. Listing users was not sufficient; each role needed a decision, a query, a quality condition and a clear failure case. This made the solution more useful and also easier to test. The next step is to interview real municipal staff, confirm their display and comparison needs, register stations, test spatial coverage, and approve averaging and alert rules before system can be used by the public.

References

UCI Machine Learning Repository. (n.d.). Beijing Multi-Site Air Quality Data. https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data
MongoDB. (n.d.-a). Schema validation. https://www.mongodb.com/docs/manual/core/schema-validation/
MongoDB. (n.d.-b). Indexes. https://www.mongodb.com/docs/manual/indexes/
World Health Organization. (2021). WHO global air quality guidelines. https://www.who.int/publications/i/item/9789240034228